ABHAY DESHMUKH

ECE STUDENT | EMBEDDED SYSTEMS & COMMUNICATION TECHNOLOGY

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PROFESSIONAL SUMMARY

Final-year Electronics & Communication Engineering student (CGPA: 8.7) with hands-on project experience in embedded systems, ESP32 sensor integration, MQTT-based data transmission, AWS IoT Core, and Python-based machine-learning applications. Interested in communication systems, data transmission, system testing, and R&D for connected and robotic systems. Brings practical experience building sensor-data pipelines and documenting technical work.

TECHNICAL SKILLS

Programming: Python, C, Embedded C
Embedded & IoT: ESP32, Arduino, sensor interfacing, MQTT, AWS IoT Core
Software & Data: Streamlit, NumPy, Pandas, TensorFlow/Keras, Plotly, GitHub
Additional: ANN/CNN classification, data preprocessing, industrial communication (Modbus, OPC UA), PLC/ladder logic

RELEVANT PROJECTS

IoT-Enabled Predictive Maintenance System

- Built an ESP32-based monitoring node using temperature, vibration (ADXL345), pressure (MPX5700AP), humidity, and noise sensors.
- Streamed sensor data to AWS IoT Core using MQTT and evaluated ANN and ensemble ML models for machine-fault classification.
- Designed an end-to-end pipeline spanning sensor acquisition, cloud data ingestion, and predictive fault detection.

Technologies: ESP32, Embedded C, Python, MQTT, AWS IoT Core, ANN

Automatic Modulation Classification Using 1-D CNN

- Developed a Streamlit app to generate noisy I/Q signals and classify BPSK, QPSK, 16-QAM, 64-QAM, CPFSK, and WBFM.
- Implemented CNN inference using TensorFlow/Keras and a pure-NumPy fallback, with interactive signal and constellation visualizations.

Technologies: Python, CNN, NumPy, TensorFlow/Keras, Streamlit, Plotly

Supply Chain Risk Classification App

- Built an interactive Streamlit prediction app with a trained classification model, feature scaling, and a 23-feature input workflow.

Technologies: Python, Pandas, Scikit-learn, Joblib, Streamlit

EXPERIENCE

Embedded Systems Intern — Pantech Prolabs India | 20 February – 20 May

- Completed hands-on training in Embedded C, microcontroller peripheral interfacing, and firmware development.

EDUCATION

B.Tech, Electronics & Communication Engineering — Tulsiramji Gaikwad-Patil College of Engineering and Technology (TGPCET), Nagpur
Expected graduation: April 2027 | CGPA: 8.7 (Final Year)
HSC: 83.60% | SSC: 89.60% — Jawahar Navodaya Vidyalaya, Yavatmal

CERTIFICATIONS & PARTICIPATION

Certifications: AWS, Python, IoT, MATLAB
Participation: Engineer's Day Celebration 2026; EngineerX – Mega Project Expo; TechCanvas – Technical Poster Competition; Industry Arena – Technical Simulation & Optimization Competition.